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RESEARCH-ARTICLE

AI Mastery May Not Be For Everyone, But AI Literacy Should Be

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AI Mastery May Not Be For Everyone, But AI Literacy Should Be

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ABSTRACT

Despite the abundance of advice from policy bodies, professional associations, advocacy groups, and scholars on how K-12 schools should assimilate AI and provide AI education, practical plans are lacking from K-12 education leaders themselves. Education leaders must make strategic decisions about how to prepare teachers and students for an AI-infused future. Simultaneously, educators need immediate support and guidance on how to manage the arrival of tools that render some existing educational practices obsolete and prompt the need to teach new skills and awareness. Near term, it may be unrealistic to expect all students to master the ability to develop AI applications; universal AI literacy is a more feasible goal. We introduce a set of short-format, modular AI literacy courses and report how they were implemented and affected teachers' and students' knowledge and perceptions of AI. Using an online questionnaire, we collected data from 265 individuals worldwide who accessed the courses, including 190 teachers who implemented them with over 11,800 students. We conducted 17 teacher interviews to gather feedback and to better understand how courses were adapted for local contexts. Teachers reported an increase in their own and their students' knowledge of AI concepts; and increased optimism about the potential benefits of AI to society and their ability to influence the future of AI. Key takeaways are that AI literacy instruction should be designed for adaptability to local contexts and cultures and that steps should be taken to institutionalize the integration of AI literacy into the regular school curriculum.

CCS CONCEPTS

• **Social and professional topics** → **Computing literacy**.

KEYWORDS

artificial intelligence, AI literacy, AI literacy curricula, K-12 education

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1 INTRODUCTION

Numerous scholars and organizations have issued definitions, curricular frameworks, policy frameworks, and a torrent of mostly untested educational opportunities about AI for students and educators. Despite the abundance of advice and resources from policy bodies such as UNESCO and the U.S. Department of Education's Office of Educational Technology [20, 21, 33]; professional associations and advocacy groups such as the Computer Science Teachers Association [CSTA], AI4K12, ISTE, Teach AI, Code.org, Common Sense Media, aiEDU, and the AI Pedagogy Project; and scholars (e.g., [18, 29, 31, 34]); we are notably lacking practical plans from K-12 education leaders and policymakers themselves on how to assimilate technological advances related to AI.

Education leaders and policymakers must make strategic decisions about how to prepare teachers and students for an AI-infused future. At the same time, educators need immediate support and guidance on how to manage the arrival of new tools that render some existing educational practices obsolete and prompt the need to teach new skills and awareness. A first step is to develop AI literacy [3, 12] among school administrators, educators, students, and their families. Long and Magerko define AI literacy as "...a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace" (p.2) [18].

The AI4K12 initiative (<https://ai4k12.org/>), a joint effort of the Association for the Advancement of Artificial Intelligence and CSTA, recently issued draft national guidelines for K-12 AI education based on five big ideas that Touretzky et al. assert every K-12 student should know about AI [31]. These guidelines aim to provide the foundation for a set of standards for education about AI and a framework for the development of curricula to teach AI concepts. In practice, despite the call for AI for all students, adoption of these guidelines will most likely be limited to computer science teachers who may have the necessary preparation and class time available. Recommendations (e.g., from Code.org, CSTA, and ECEP Alliance) that instruction about AI be embedded in mandatory computer science courses for all students are similarly unattainable near-term goals given that, across 35 states with relevant data, only 5.8 percent of high-school students are enrolled in foundational computer science courses [35].

Some level of knowledge about and mastery of AI should be considered a basic skill that everyone should acquire—akin to universal literacy. Long term, the goal may be to offer every student the opportunity to master the programming skills needed to modify and create AI applications. For now, both teachers and students need to be sufficiently AI-literate to be savvy users of AI: understanding what AI is, how it is used, and how it can help or harm society.

1.1 Foundations for AI Literacy Instruction

Schools can take immediate steps to develop AI literacy using instructional materials that address Touretzky et al.'s five big ideas [31]. The AI4K12 curriculum guidelines enumerate relevant concepts related to each of the five big ideas and provide a progression of learning objectives and “enduring understandings” by grade band. Long and Magerko developed a conceptual framework for AI literacy comprising five overarching themes: “What is AI?; What can AI do?; How does AI work?; How should AI be used?; and How do people perceive AI?” (p.3) [18]. The AI4K12 guidelines and Long and Magerko’s framework establish a foundation from which AI literacy curricula can be developed and evaluated.

Numerous efforts to develop AI-related educational experiences and curricula for students have yielded practical design considerations. Williams et al. recommend that instructional materials on AI concepts should incorporate active learning and “unplugged” (p.3) activities, engage students in projects, and help students consider the ethical implications of AI applications [37]. They highlight the need to “mind the digital gap” (p.41) and lower barriers to entry by limiting the amount of prerequisite knowledge and technology resources students need to engage in the content. Lin and Van Brummelen emphasize the importance of co-designing AI educational experiences with practicing teachers and integrating AI concepts into core subjects to improve accessibility to both teachers and students [17]. Similarly, Sabuncuoglu documents the value of incorporating teacher and student concerns and feedback in the design process [25]. Ng et al.’s comprehensive review of AI education over a 20-year period observes that the arrival of tools such as robotics kits, intelligent agents, and interactive platforms that minimize the need for technical knowledge and coding skills have greatly facilitated the teaching of AI concepts [23]. They also note the value of collaborative projects, project-based learning, and game elements for engaging students in AI education. Laupichler et al. highlight the need to evaluate courses to ensure accountability and appropriate teaching practices [15].

Initial efforts to implement and evaluate AI literacy instruction for students have been promising. Reviews by Ng et al. and Su et al. found that researchers studying AI literacy efforts employed a variety of data collection methods to gather data on cognitive and non-cognitive constructs related to AI [24, 29]. These included knowledge assessments to assess student abilities and knowledge gains related to AI; questionnaires to gather perceptions of cognitive abilities, usefulness of AI, and optimism with respect to AI; interviews, observations, and artifact analysis. In the US, studies have generally been small-scale with self-selected students in out-of-school settings (e.g., [8, 16, 37]). Additionally, the researcher-developers themselves taught or participated in teaching the content and administered the assessments of impact on students. Larger-scale

implementations and evaluations of AI literacy instruction have taken place in Hong Kong [5, 14] where there is a concerted effort to create a territory-wide pretertiary AI curriculum and to develop and validate instruments to assess AI literacy. If the goal is for all students to be AI literate, AI literacy will need to be fostered by schoolteachers in regular school settings. Furthermore, as AI is a new topic to many teachers, they must be adequately prepared to provide AI instruction to a population of students that is diverse in culture and academic preparation. A precondition is building teachers’ trust in AI as a useful tool to complement rather than replace their work [13, 22]. Coordinated efforts to institutionalize AI literacy, similar to Hong Kong’s, are needed in the US and elsewhere.

1.2 The Day of AI Courses

Building on these foundations and to meet the surging need for widely-usable AI literacy instructional materials, over the past two years, Massachusetts Institute of Technology and an implementation partner, i2Learning, developed 12 modular, short-format AI literacy courses with an emphasis on responsible, equitable, and ethical design and use of AI (details in Table 1). Three derivative courses were created for educators in Australia. The course design incorporates general principles of effective literacy instruction [1, 11, 27] including introducing and defining vocabulary necessary to understand AI concepts; activating students’ prior knowledge and experiences; providing relatable examples of AI and what is not AI; using the new vocabulary in explaining how AI works; incorporating videos and slide decks to provide context and visual cues; engaging students in student-centered, hands-on, collaborative activities that apply AI concepts; providing opportunities for students to create meaningful products; and engaging students in small-group discussions about the ethics and responsible use of AI, issues of equity, and how AI may benefit or harm individuals and society. The courses address concepts from each of Touretzky et al.’s five big ideas in AI and incorporate the National Research Council’s core ideas of engineering design [6, 31]. However, while Touretzky and Gardner-McCune imply that the AI4K12 guidelines aim to integrate AI literacy into broader instructional programs on computer science or computational thinking [32], these instructional materials are intended to provide a standalone introduction to AI literacy for immediate implementation in schools. Most require no student or teacher coding experience and can be taught by teachers of any disciplinary background.

The Day of AI materials, including educator guides, teacher slides, student resources, and training videos, are freely available online at <https://dayofai.org/>. They can be downloaded, adapted, and redistributed under a Creative Commons license and are usable by educators with no experience teaching AI and with little equipment beyond digital devices and internet access. Each course consists of 1-9 lessons that can each fit into a typical lesson period or be delivered over the course of 1-2 days dedicated to AI literacy. A Spanish translation has been available since 2023. Between February and July 2023, approximately 7,600 people, mostly teachers, in 136 countries registered to access the Day of AI materials.

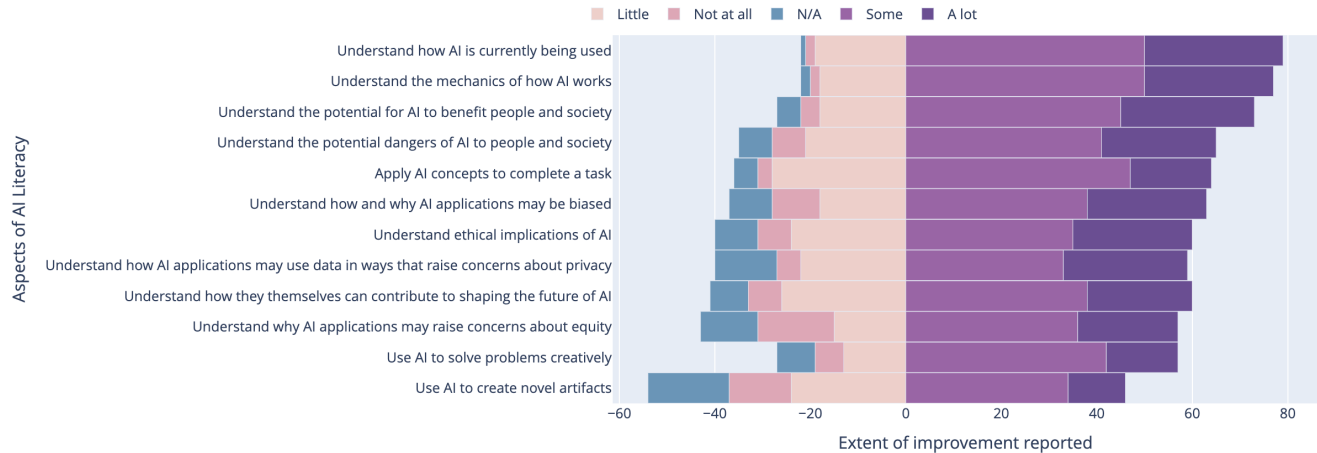


Figure 1: Teachers' Ratings of the Extent to Which the Courses Improved Specific Aspects of Students' AI Literacy

2 METHODS

We gathered data on the Day of AI courses using a voluntary online questionnaire and virtual interviews. The study was deemed exempt from human subjects research review. The questionnaire was built in Qualtrics and included multiple choice, dropdown, sliding scale, and open-ended questions. All respondents were asked how the course content affected their views of AI and knowledge level about AI, and for demographic information. Those who implemented the Day of AI courses with students (“implementers”) were asked a more extensive set of questions to elicit information about the types of students to whom they taught the content, impact on their students, and to ask for content-specific feedback. Implementers were also asked if they were open to a follow-up interview. Invitations to complete the online questionnaire were emailed four times in May and June 2023 to all 7,204 individuals who registered to access the courses and also provided a valid email address for contact. Seventy-five respondents answered the first set of questions only with a median completion time of 6 minutes. One hundred and ninety implementers answered both sets of questions with a median completion time of 16 minutes. Responses to the questionnaire’s close-ended questions were analyzed using Excel and SPSS to summarize the data in frequency tables and to calculate descriptive statistics where appropriate. Results reported as percentages are generally based on the number of responses to the relevant question. Responses to each open-ended question were coded and themed using a general inductive approach [30]. Following Creswell, inductive coding involved a close reading of each respondent’s text response to identify segments directly relevant to the question and evaluation objectives [7]. Each segment was assigned a key word or phrase to categorize it. Redundant or overlapping categories were combined to reduce the overall number of categories and develop overarching themes reflecting the key ideas raised by respondents.

A structured interview protocol [19] elicited additional information and feedback from 17 implementers who volunteered to be

contacted for an interview. They were asked how they implemented the courses, any obstacles encountered, and adaptations needed for the local context. Interviews lasted 34–65 minutes and, in most cases, the interviewee gave permission for the interview to be recorded and transcribed. Ten of the interviewees were located in the US and seven interviewees were from abroad: Spain (two interviewees), Algeria, Scotland, Guatemala, Singapore, and Australia. Author 1 conducted a content analysis of the interview notes, identifying key themes that emerged [36].

3 FINDINGS

3.1 Settings and Participants

Questionnaire respondents were located in 38 countries with 68 percent based in the US. The majority of respondents were female, White or Asian, over 35, and had earned a graduate degree. Respondents held degrees in a range of disciplines but the most common were education, computer science, science, and technology. Ninety-one percent of the implementers were teachers or instructors and 6 percent were curriculum coaches. Implementers’ teaching experience ranged from less than a year to 60 years, averaging 16 years. They taught a wide range of subjects including social studies, literature, languages, the arts, and special education but the most common were computer science, technology, mathematics, science, and engineering. Ninety-five percent ($n=151$) of implementers taught general education students, 49 percent taught students with disabilities, and 40 percent taught students who are not fluent in the primary local language. Implementers scored a mean of 9.26 on a scale of 0–10 indicating their level of comfort using computers (0 = not at all; 10 = very). Almost two-thirds of the course implementations were in government-funded schools. Twenty-two of the American schools in which the courses were taught were Title I schools. Implementers collectively taught the courses to at least 11,884 individuals aged from under 5 to over 50 years old (including teachers and other staff members). Over half the implementations were with students of mixed ages. The majority of them reached

Table 1: Day of AI Courses

Course	Target Age (yrs)	Topics addressed
What Can AI Do?	5-7	Define AI; identify AI applications, capabilities, limitations; AI sensors; image recognition and sorting
Teachable Machines	8-10	Machine learning: collecting datasets, training an algorithm, predicting on new data, algorithmic bias
AI Blueprint Bill of Rights	8-10	How human rights are impacted by the use of AI: non-discrimination, privacy, transparency, safety
Can Machines Be Creative?	11-13	Machine learning: collecting datasets, training an algorithm, predicting on new data; algorithms; generative adversarial networks; deepfakes and data manipulation; ethical and responsible use of AI
Game AI	11-13	Introduction to Scratch, rule-based learning vs. reinforcement learning and ethical implications
AI in Social Media	11-18	AI in recommendation systems, clustering algorithms, and filter bubbles; how misinformation can spread; responsible use and regulation of social platforms
Data Science and Me (no coding)	14-18	Creating datasets, sharing data, data activism, bias in data, data visualizations, how identity and power can be expressed through data
Data Science and Me (with coding)	14-18	Creating datasets, aggregating and cleaning data, combining data sources using Python, data visualizations, how data are captured about individuals by governmental and social technology systems, sharing data, how identity and power can be expressed through data
Intro to Voice AI	14-18	How voice AI works, programming Alexa skills using voice AI, how Alexa uses AI to determine what user commands mean
Personal Image Classifier	14-18	Image classification, using App Inventor to create a personal image classification model, bias in machine learning systems, ethical implications of using AI in image recognition
Data Science and Decision Making	14-18	AI in college admissions decisions, biased AI and algorithmic decision making, analyzing and visualizing data, implications of using and not using affirmative action policies
ChatGPT in School	8-18	AI and the creative process, how large language models work, chatbots vs. search engines, benefits and limitations of large language models, how they should be used in schools

students aged 9-18 years: 41 percent included elementary school students, 68 percent included middle school students, and 63 percent included high school students. The average instructional time spent on a course was 4.3 hours, and ranged from 2.3 to 8.5 hours.

3.2 Survey and Interview Results

3.2.1 What Teachers and Students are Learning as a Result of Engaging with the AI Literacy Courses. All respondents were asked to indicate on a scale of 0-10 their level of knowledge about AI before and after engaging with the course content (0 = no knowledge; 10 = expert knowledge). A paired samples t-test found a significant difference in self-reported level of knowledge from before ($M=3.8$, $SD=2.03$) to after engaging with the content ($M=6.0$, $SD=2.22$); $t(257)=20.936$, $p<.001$. Implementers were similarly asked to rate their students' level of knowledge about AI before and after implementing the courses. The weighted average score across courses increased from 2.4 to 4.3. Implementers also rated the extent to which the courses improved specific aspects of their students' AI literacy. Results are shown in Figure 1.

3.2.2 Impact on Teacher Perceptions of AI and Their Sense of Agency in Determining its Place in Their Future. All questionnaire respondents were asked to indicate on a scale of 0-10 their level of optimism about the potential benefits of AI to society before and after engaging with the courses (0 = not at all optimistic; 10 = very optimistic). A paired samples t-test found a significant increase in score from before ($M=5.6$, $SD=2.39$) to after ($M=7.1$, $SD=2.10$); $t(256)=11.005$, $p<.001$. Common themes that arose in open-ended responses explaining how the courses increased respondents' optimism about the potential benefits of AI to society included learning about practical ways in which AI is already being used in daily life to improve the accuracy and efficiency of tasks, especially in medicine and in education; learning how AI can help people with special needs; and the emphasis on ethical and responsible use of AI. Issues that reduced optimism included the speed at which AI is evolving, the lack of governance about how it is used, misuse of personal data, bias, misinformation, and concerns about human-computer relations.

Questionnaire respondents varied in the extent to which they felt they could contribute to shaping the future of AI but were generally more positive about their ability to contribute to shaping the future of AI literacy (defined in the questionnaire as "people's ability to understand the role of AI in our lives, to evaluate its pros and cons, and to use it productively"). Learning more about AI gave 63 percent ($n=242$) of the questionnaire respondents more confidence in their ability to contribute to its future. The majority of these respondents felt they could exert influence by teaching their students about AI. For example, a teacher in a district school in Alabama wrote "As an educator, I feel that exposing the students to AI, teaching them the basics of AI and addressing the social concerns is a vital part of shaping the future of AI. These students will be the ones to be the main contributors to the growth and expansion of AI." Encouragingly, only 1 percent of the respondents reported that learning more about AI decreased their confidence in their ability to contribute to the future of AI. A multiracial, female teacher in California working in a remote urban public school expressed her concerns as follows: "Understanding more about what AI is, how it

is being used and who is in control of the train concerns me because the decision-makers do NOT share my values.”

3.2.3 Impact on Student Perceptions of AI and Their Sense of Agency in Determining its Place in Their Future. Implementers were asked to rate their students’ level of optimism about the potential benefits of AI to society before and after they had implemented the course. The weighted average score across courses increased from 5.0 to 6.7. Sixty-one percent of implementers reported that their students felt they can contribute a lot or to some extent to shaping the future of AI. Seventy-seven percent (n=151) of implementers reported that, as students learned more about AI, they felt more able to contribute to its future. Some explained that students had begun to understand how their own data are being used in AI models and that their use of AI applications might contribute to training the models. Other implementers suggested that students saw a more active future role for themselves in developing AI applications. For example, a teacher in New York who taught “Teachable Machines” to one hundred 9-12-year-olds asserted that: “Students are understanding how AI is being used and created and they are understanding that they can become programmers and creators of this technology and not just passive users of it.” For a few students, that moment of control had already arrived: a teacher in North Carolina reported “I had one student who leapt out of their seat when they tested their Teachable Machine model and it worked! They were using it to assess the ripeness of produce and envisioned a future in which robots would do our grocery shopping...”. Twenty-three percent of the implementers indicated that students’ feelings about their ability to shape the future of AI had not changed, some noting that it would take more time on the topic to have an impact.

3.2.4 Cultural Relevance, Equity, and Social Justice. Implementers rated the courses on a scale of 0-10 on how well they addressed issues of equity and social justice, and on how culturally relevant they were for their students (0 = not at all; 10 = very well/very). The mean weighted scores across the 15 courses were: 7.0 for cultural relevance (range 6.0 – 10.0), 6.1 for equity (range 3.0-10.0); and 6.0 for social justice (range 2.7 to 10). Culturally relevant content was defined for respondents as that which “affirms and validates students’ cultural identities and experiences; is inclusive and representative of diverse perspectives and cultures.” Neither equity nor social justice were defined which may have resulted in different understandings of these questions. Respondents’ written comments suggested that many were unclear on the meaning of social justice and how it might apply to instructional content.

3.2.5 How Can the AI Literacy Activities be Improved to Actively Engage More Teachers and Students? The majority of implementers found the educator/teacher guide, tutorial videos, teacher slides, and student resources very helpful and appreciated being able to adapt them for their own contexts. The ready-to-use teacher slides were the most well-received resource with 79 percent of the implementers (n=170) rating them as very helpful. Suggestions for improving the courses most often related to ways of minimizing didactic approaches to teaching in order to limit the amount of time the teacher is instructing and increase time spent by students engaging with the materials and AI-based applications. These suggestions included providing more opportunities for interaction and

Q&A with students, for example, including quizzes, and using tools such as Pear Deck, Curipod, or Padlet; offering more advanced content and extensions; increasing the number of games or game-like activities, videos, video tutorial explanations, real-world examples, hands-on activities, project-based learning activities, opportunities for students to explore concepts and content to discover for themselves, and independent activities. Other suggestions for improvements focused on increasing the accessibility of the materials to more educators and students. These included providing differentiated training for teachers of varying disciplinary backgrounds, improving accessibility of the content and applications for people with disabilities, and translations into other languages.

Implementers were asked for suggestions as to how the courses could be made more culturally relevant for their students. Responses included introducing AI and its potential benefits to parents who may act as gatekeepers preventing their children from using and learning about it; incorporating voice activation tools that can recognize different accents or ways of speaking; using social media popular with teenagers from different racial/ethnic groups; using applications in which students can customize the digital environment to make it relevant to themselves; showcasing individuals, characters, and data representing people from different backgrounds; focusing on how AI can help with English Language Learners; introducing AI activism; and including activities that allow students to use AI-based tools for a typical school task or assignment.

Implementers were also asked for suggestions as to how the courses could better address issues related to equity. Suggestions included applying principles of Universal Design for Learning to improve equity of access to the content; including examples of AI-based gadgets or machines that are more affordable to low-income families; discussing the digital divide and to what extent AI applications are available in different regions and countries, and possible consequences for those who do not have access; and providing information about bias that affects people with disabilities and how AI can support people with disabilities.

Finally, implementers were asked for suggestions as to how the courses could better address issues related to social justice. Ideas included adding more current, real-life examples of bias in AI, its dangers, and how to apply the concept of social justice to AI; allowing students to explore solutions to bias in AI; providing examples of solutions to bias in AI; providing more information about copyright and credit or attribution issues related to AI-generated content, for example, for datasets being used; and discussing AI and human decision-making, and how responsibilities are shared.

4 DISCUSSION

At the outset of this article, we suggested that aiming for universal AI literacy might be a more realistic near-term goal than AI mastery. We subsequently described a large-scale, global implementation of AI literacy instruction taught under routine conditions by teachers from many disciplines. The results of our investigation into the implementation and perceived effects of this instruction provides a promising indication that, even with very short exposure to AI literacy activities, both educators and students in a wide range of contexts can gain knowledge about how AI works, key AI concepts, current uses of AI, and potential benefits and harms to society. In

addition, learning more about AI increased their levels of optimism about the potential benefits of AI to society and about their own abilities to contribute to shaping the future of AI. The study points to ways in which more formal AI literacy instruction can be developed and improved to engage more students. It also provides ideas about how to institutionalize AI literacy to ensure that all students are, borrowing from Bawden’s definition of literacy, able to “deal with information using the technologies and formats of the time” (p.18) [2].

4.1 Improving AI Literacy Instruction

Teachers offered many suggestions for improving student engagement in the course content and for increasing its accessibility, cultural relevance, and attention to issues of equity and social justice. Clearly, no single course or curriculum will be ideal in every context making it imperative that AI literacy instructional materials should be designed to facilitate local adaptations to integrate with existing curricula and tools, and to reflect local context, academic standards, practical implementation constraints, culture, and values [4, 9, 26]. The instructional materials must also be regularly updated to reflect the latest AI developments and tools. Professional learning opportunities for teachers should provide differentiated training and supports for teachers of varying background knowledge and skills. Furthermore, to ensure broad appeal, AI literacy instruction for students and professional learning for teachers should help teachers and students understand how AI, within local policy guidelines, can be useful in executing typical school-related tasks and assignments. Curricula targeting low-income students, including Title I schools, could integrate basic literacy and math skills with AI literacy activities.

While AI literacy instruction developed to date has often been based on design and competency frameworks and guidelines, development of more specific AI literacy standards and of validated assessments that are usable across a variety of AI literacy curricula would ensure greater alignment between intended, enacted, and assessed curricula [10]. They would also allow for more rigorous evaluation of instructional materials and curricula to determine which aspects of AI literacy they are able to improve and which aspects need further emphasis or different design features. To date, assessments used for AI literacy have been created by researchers for specific studies (see [24, 29]) but have not been tested in routine educational settings with large and diverse populations of students. AI literacy standards would help ensure AI literacy curricula meet expectations for high-quality educational resources while broadly applicable, validated assessments could be used in impact evaluations of AI literacy curricula to determine whether students are indeed becoming more AI literate.

4.2 Institutionalizing AI Literacy

The data we collected on characteristics of the teachers who implemented the Day of AI courses, and on how and why they did so, suggest that, in most educational settings, AI literacy is still an optional topic being adopted by first movers. While the courses were taught by some teachers without a science background, most implementers taught computer science, technology, math or science. Regardless of discipline, the majority were highly educated,

experienced, and very comfortable using computers. Over half of them took time to adapt the instructional content for their local contexts. In addition, most of these educators appear to have made the decision to implement the Day of AI courses without urging from administrators or parents. Further evidence suggesting that implementers were mostly early adopters is that, for almost three-quarters of the implementations, the teacher was the only one at their school implementing the courses. More widespread adoption of AI literacy curricula will require attention to educators’ cognitive, emotional, and contextual concerns, including the provision of institutional support [28].

To extend the reach of AI literacy instruction beyond highly-motivated, self-selecting educators and their students, institutional adoption is needed. While it is important for teachers to feel they have a choice in adopting new technologies [28], decisions to introduce AI literacy should be made at the school or district level. This will require educating decision-makers such as school board members, superintendents, chief technology officers, and school principals about the pros and cons of training teachers and teaching students about AI, and helping them develop local policies governing the use of AI in the schools they oversee. Professional associations may play a key role in reaching these decision-makers while federal, state, and local education departments can establish policy guidelines to be adapted for local contexts. Even parents must be educated about why their children should learn about AI given their role as gatekeepers of their children’s educational opportunities.

4.3 Study Limitations and Future Directions

Confidence in the study findings are limited due to Day of AI teachers being self-selected and due to the self-report nature of the questionnaire and interview data. Additionally, the study relies on teacher reports of student learning and changes in attitudes rather than measuring student outcomes directly. Nevertheless, the results are impressive given how little training teachers were provided and the short amount of time students engaged in the AI literacy content relative to other topics and subjects they study. Future studies of AI literacy programs should include pre- and post-course assessments for a treatment and comparison group of students, ideally randomly assigned, to more rigorously assess learning and changes in beliefs. Studies that investigate whether AI literacy programs effectively address issues of equity and social justice should clearly define the terms to ensure a common understanding among study participants.

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